Kiron Sukesan

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Summary

Software and machine learning engineer with 9+ years of experience spanning ML systems, distributed infrastructure, and biomedical signal processing. Track record building large-scale production systems at Amazon and ML platforms at Compass. Research background in Bayesian probabilistic modeling and adaptive signal acquisition with peer-reviewed publications. Targeting senior ML/research engineering roles at the intersection of applied ML and systems.

Skills

Languages	Java, Python, C++, Rust, JavaScript, MATLAB, Bash
ML / AI	ANN, HNSW, Computer Vision, Gaussian Processes, Bayesian Inference, Semi-Supervised Learning, Deep Learning, Reinforcement Learning
Storage Systems	Distributed Database Internals, Consensus Protocols (Paxos, Raft), Database Engine Implementation (LSM tree and B-tree)
Tools & Infra	AWS , Docker, Git, Linux

Experience

Software Development Manager *at* **Amazon Web Services (AWS)** Aug 2024 – Present
DynamoDB Storage Node, Seattle, WA

- Led part of the engineering organization for DynamoDB's Storage Node, one of AWS's highest-scale distributed storage systems, overseeing architecture, delivery, and team health. Specifically responsible for the TTL system, replication protocol, general correctness of DynamoDB Storage, and feature expansion efforts within storage.
- Drove cross-functional alignment across engineering, product, and operations to execute a multi-quarter roadmap spanning reliability, performance, and new feature delivery.
- Responsible for hiring, mentoring, and performance management for a team of 10+ junior, mid-level and senior software engineers; established engineering standards and oncall culture.

Senior Software Engineer *at* **Amazon Web Services (AWS)** Jan 2023 – Aug 2024
DynamoDB Storage Node API Team, Seattle, WA

- Designed and implemented features for the DynamoDB Storage Node API layer, contributing to a system serving millions of customers with stringent latency, availability, and durability requirements.
- Collaborated with principal engineers on storage node architecture to improve reliability, latency, and throughput for core DynamoDB operations at global scale.
- Led oncall and operational excellence initiatives, driving root cause analyses and systemic improvements to reduce customer-impacting events across the storage tier.

Machine Learning Engineer II / Senior Machine Learning Engineer *at* **Compass** Dec 2019 – Nov 2022
Seattle, WA

- Built and owned Computer Vision based ML models used by agents across the platform, directly influencing transaction outcomes.
- Architected end-to-end ML pipelines spanning data ingestion, feature engineering, model training, evaluation, and production deployment on cloud infrastructure.
- Mentored junior ML engineers and drove platform improvements that reduced model iteration cycle time and increased experiment throughput.

Software Development Engineer I/II *at* **Amazon Kindle** Oct 2017 – Nov 2019
Greater Seattle Area

- Designed and shipped features for Amazon Kindle's internal data processing pipeline, collaborating with senior engineers on distributed systems design, API contracts, and service reliability.
- Participated in oncall rotations and drove root cause analyses for production incidents, contributing to improved monitoring and alerting infrastructure.

- Promoted to SDE II after demonstrating technical ability with end to end solution for Kindle compliance with GDPR's requirement to return data to customers on request.

Data Scientist / Research Engineer *at* **Washington University in St. Louis** May 2016 – Aug 2017
Laboratory of Sensory Neuroscience and Neuroengineering, St. Louis, MO

- Developed a Bayesian active learning framework using Gaussian Process Classification to efficiently estimate audiometric threshold surfaces, reducing clinical measurement burden while preserving diagnostic accuracy.
- Extended the framework to extract latent hearing loss features undetectable by standard audiometry, enabling potential early-stage diagnosis prior to clinical threshold shifts.
- Implemented a scalable .NET web application for real-time audiogram generation, deployed for ongoing comparative research between ML-based and standard audiology methods.
- Designed a novel ML-based hardware calibration algorithm for audiometric equipment, improving signal fidelity and measurement consistency across experimental sessions.
- Co-authored two peer-reviewed publications in psychometric field estimation and Bayesian audiometry (see Publications).

Software Developer *at* **Purnatva Solutions Pvt. Ltd.** Sep 2014 – Aug 2015
Bengaluru, India

- Implemented real-time GPS bus tracking module for the Falcon Mobile App, supporting large-scale deployment with 10,000+ active users across the platform.
- Contributed to *vidyadhan.org*, a LAMP-stack scholarship portal connecting sponsors to underprivileged students, working directly with the product team on feature delivery and backend development.

Biomedical Engineer, Undergraduate Research *at* **Wash. U. in St. Louis** Aug 2013 – May 2014
Department of Biomedical Engineering

- Designed and built a wearable EEG recording device for overnight clinical use, using custom analog circuitry, embedded software, and mechanical design in close collaboration with two biomedical engineering peers and clinical partners.
- Validated device performance against clinical EEG standards and iterated on hardware and firmware based on clinician feedback.

Neurosurgery Research Intern *at* **University of Washington** Jun 2013 – Aug 2013
Brain-Computer Interface Lab (Dr. Jeff Ojemann), Seattle, WA

- Developed a signal analysis algorithm for electrocorticographic (ECoG) neural signals; diagnosed and resolved electromagnetic noise artifacts that had been corrupting recorded data.
- Constructed a closed-loop BCI device that enabled motor-impaired patient hand movement through real-time classification of theta wave activity from ECoG recordings.

Education

Washington University in St. Louis 2015 – 2016
Master of Science, Computer Science St. Louis, MO

Washington University in St. Louis 2010 – 2014
Bachelor of Science, Biomedical Engineering St. Louis, MO

Publications

- **Sukesan, K.** et al. "Bayesian active probabilistic classification for psychometric field estimation."
- **Sukesan, K.** et al. "Conjoint psychometric field estimation for bilateral audiometry."